Detection of the Status of Diatom Blooms in the Tributaries of the Yangtze River Based on Sentinel-2 Images

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***Abstract*— Diatom blooms frequently observed in the river tributaries pose a threat to the regional water environment. Satellite remote sensing provides us with an effective tool to delineate the extent of large-scale harmful algal blooms (HABs) in both oceans and inland waters. However, the diatom bloom detection in river systems remains challenging, due to the high demand for a robust model to characterize the spatial and spectral features from the satellite images with high resolution and limited spectral bands. In this study, we developed a novel deep learning-based framework for the detection of the status of diatom blooms in river tributaries using Sentinel-2 MultiSpectral Imager (MSI) images. Distinct from the previous works for detection of bloom extent, the water pixels are categorized as "nonbloom," "mild bloom," and "severe bloom," corresponding to the different bloom intensities. To achieve this, a simple convolutional neural network (CNN) is trained using the collected spectral samples from MSI images characterizing the different bloom statuses. The input features include the spectral variables highly correlated to the pivotal absorption and backscattering properties of diatoms, chlorophyll-a (Chla), and water temperature. The classification results can then be obtained based on the spectral and environmental characteristics learned by the constructed model. The trained model was extensively tested in tributaries of Yangtze River, where both visual and quantitative validation demonstrated that the proposed model can achieve reliable detection results.**

***Index Terms*— Deep learning, diatom bloom, remote sensing, river tributaries, Sentinel-2.**

I. Introduction

ABOUT two-thirds of the Earth's surface is covered by water; however, less than 0.01% of the freshwater is directly available to human beings [1]. In recent decades, global warming and intensified urbanization have brought new threats to the sustainable utilization of freshwater resources. As a result, harmful algal blooms (HABs) frequently occur, associated with the increasing eutrophication of water, and have an adverse impact on the aquatic environment [2].

In addition to the scattered in situ ground sites for monitoring water quality with laboratory analysis, remote sensing provides us with a complementary tool to delineate algal bloom extents and dynamics in a cost-effective way [3]. Generally speaking, the water-leaving radiance measured by the satellite sensor incorporates the scattering and absorption characteristics of the dissolved and suspended constituents of water [4]. Therefore, algal bloom reflectance within the near-surface water column is distinguishable from clean water based on the spectral characteristics. Algal blooms can be composed of various phytoplankton species, e.g., blue-green algae, diatoms, and Phaeophyta [5]. Accordingly, the absorption spectra vary with the dominant photoactive pigments within the water, which produce unique spectral reflectance that can be measured by satellite sensors [5]. Based on these facts, the analysis of algal taxonomy is feasible using remote sensing images with fine-resolution spectral bands covering the featured wavelengths [6].

Over the past decades, the detection of HABs based on remote sensing has drawn a lot of attention, with a variety of methods being developed. One of the popular working pipelines is to design a spectral index to capture the discriminative characteristics of algal blooms. Examples are the floating algae index (FAI) using the red/near-infrared (R/NIR)/short-wave infrared (SWIR) associated feature for the detection of floating algae in open ocean environments [7], the red tide detection index (RDI) using a three-band combination (667, 555, and 748 nm) for detecting algal blooms in turbid coastal waters [8], the maximum chlorophyll index (MCI) [9], and other similar R/NIR algorithms for algal bloom detection over turbid eutrophic inland waters [10], [11]. The index-based algorithms are usually easy to implement and computationally efficient. However, the indices are constructed based on the narrow feature bands associated with different satellite sensors and are difficult to apply to multispectral sensors with a relatively coarse spectral resolution. Moreover, the optimal thresholds for the indices can vary in different regions, which is another challenge when adaptively utilizing these indices in large-scale applications.

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The supervised methods generally involve samples acquired from manual interpretation or the existing datasets and can show robust performance in the task of algal bloom detection [12]. As a popular way, the machine learning classification methods use labeled bloom and nonbloom samples to train the classification model and generate the detection results [13], [14]. More recently, researchers have made attempts at developing deep learning-based algal bloom detection methods, where the classical convolutional neural network (CNN) [15] and U-Net models [16] have been applied to discriminate various HABs in oceans and coastal waters. The models established the nonlinear correlation between the input spatial and spectral features and the output labels and achieved promising results with the powerful capabilities of deep learning for feature learning and discrimination. In addition to the detection of the spatial extent of algal bloom area, the data-driven approaches using machine learning and deep learning models have also shown great potential to discriminate different algal species. By learning the distinctive spectral features implicitly contained in the input for training the classification model, this category of method can achieve a robust performance and is adaptable to the different sensors [17], [18].

Although a large number of studies have been conducted for algal bloom detection, the majority of the studies have paid attention to the HABs occurring in marine systems and inland lakes [3]. Since the late 1990s, ocean color satellite sensors with narrow wavelength bands have provided abundant images for the identification of algal blooms in open ocean and coastal areas, including SeaWiFS, Terra/Aqua MODIS, the Medium Resolution Imaging Spectrometer (MERIS), and the new-generation Ocean and Land Color Instrument (OLCI). However, the coarse spatial resolution of these sensors (300–500 m) significantly limits their monitoring capability for inland waters. In recent works, the use of medium-resolution satellite images (e.g., Landsat8 and Sentinel-2) for delineating algal bloom events in inland waterbodies has been explored over global scales [3], [10]. Nevertheless, the main targets of these studies have been the blue-green algal blooms dominant in eutrophic freshwater lakes and reservoirs, whereas the harmful diatom blooms potentially existing in fluvial environments have drawn little attention. Although algal blooms rarely occur in the main river channels, the tributaries flowing through urban regions at a low velocity and with nutrient discharges are hotbeds for the harmful species of phytoplankton, which pose a threat to the regional water environment that cannot be ignored.

The algal bloom detection in river systems has a high demand for a robust model to characterize the spatial and spectral features from the remote sensing images with a high resolution and limited spectral bands. To this end, we aimed to develop a novel deep learning-based method for the accurate detection of the status of diatom blooms in river tributaries using Sentinel-2 MultiSpectral Imager (MSI) images. The Han River, which is one of the most nitrogen-contaminated tributaries of the Yangtze River in China [19], was selected as the study area. Distinct from the previous works for extraction of bloom extent, a simple CNN model was trained for the classification of the statuses of diatom blooms, which can provide critical information for early warning for diatom blooms in rivers. Specifically, the water pixels were categorized as "nonbloom," "mild bloom," and "severe bloom," based on the spectral and environmental characteristics learned by the model established in a data-driven manner. Finally, the performance of the proposed method was validated both qualitatively and quantitatively, and the random forest (RF) model was applied for comparison. The primary goal of this study was to propose a new framework to monitor the diatom blooms occurring in river tributaries, based on broadband remote sensing images, and to provide the detailed spatial and temporal information about the diatom blooms required for effective pollution prevention.

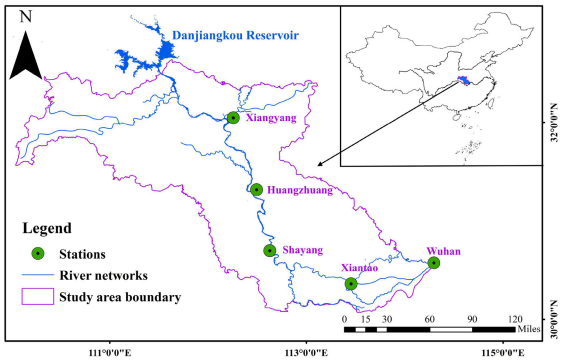


Fig. 1. Location of the middle and lower reaches of the Han River.

II. Study Area and Data

A. Study Area

The Han River is one of the major tributaries of the Yangtze River in China. The study area covers the middle and lower reaches of the Han River, with a channel length of ∼652 km [20]. The climate in this area is controlled with a subtropical monsoon mode and is characterized with mild temperature and low precipitation in winter. The downstream part of the Han River flows though several big cities with dense populations in Hubei province, as shown in Fig. 1. The discharge of industrial and agricultural wastewater into the river course and also domestic sewage are causing pollution issues for the river system. As a result, algal blooms have frequently occurred in the Han River since 1992, where brown-colored diatoms (i.e., *Cyclotella* and *Stephanodiscus hantzschii*) are the dominant algal species. The diatom blooms usually occur in the late winter and early spring, associated with a water temperature of 2 °C–20 °C and a low flow velocity [21].

B. Satellite Data

In this study, the main data source was Sentinel-2 L1C data obtained from the Copernicus Open Access Hub (https://scihub.copernicus.eu). The MSI sensor onboard Sentinel-2A/B collects optical images with 13 bands covering the wavelengths from visible and NIR (VNIR) to SWIR. The spatial resolution of the MSI images differs according to the band wavelength. In this study, we used the four 10-m bands and five 20-m bands for the monitoring of diatom blooms

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TABLE I
DETAILED INFORMATION ABOUT THE SENTINEL-2A MSI BANDS USED IN THIS STUDY

Satellite /Sensor	Band	Central wavelength (nm)	Bandwidth (nm)	Spatial resolution
Sentinel-2 MSI	B2-Blue	492.7	65	10 m
	B3-Green	559.8	35	10 m
	B4-Red	664.6	30	10 m
	B5-Red edge	704.1	14	20 m
	B6-Red edge	740.5	14	20 m
	B7-Red edge	782.8	19	20 m
	B8-NIR	832.8	105	10 m
	B8A-Red edge	864.7	21	20 m
	B11-SWIR	1613.7	90	20 m

in rivers. The detailed information about the MSI bands is presented in Table I. The revisit frequency of each Sentinel-2 satellite is ten days, and the joint revisit frequency of the A/B satellites reaches five days. In this study, 34 images with less than 30% cloud cover acquired during 2017–2021 in the study area were downloaded, where 26 images were used for constructing the training set to establish the diatom bloom detection model and the remaining eight images were used for testing the model performance.

The land surface temperature (LST) product of the Sentinel-3 satellite was also used to provide daily water temperature with a spatial resolution of 1 km as an auxiliary clue to identify the diatom bloom pixels. The Sentinel-3 LST product was generated from the data acquired by the Land and Sea Surface Thermometer (SLSTR) onboard the Sentinel satellites using the split-window algorithm [22].

C. In Situ Data

In situ data were collected in the middle and lower reaches of the Han River by the Ecology and Environment Monitoring and Scientific Research Center (EEMSC) of the Ministry of Ecology and Environment of the People's Republic of China. The historical records of the in situ surveying for algae blooms in the study area from 2017 to 2021 used in this study are presented in Table II. As shown in Table II, algal blooms have been frequently observed in the mainstream area between Huangzhuang and the station of Wuhan in the early spring during the past years, where the survey results indicate that *Stephanodiscus hantzschii* and *Cyclotella* belonging to diatoms are the dominant species. The survey records are used for collecting the training and test samples, where the diatom bloom statuses for the images were judged based on the representative water quality indicators, i.e., chlorophyll-a (Chla) concentration and algal cell density (ACD).

Moreover, in situ hyperspectral $R_{rs}(N_{R_{rs}} = 180)$ was used to evaluate the quality of the atmospheric correction result (see the Appendix). For the construction of the machine learning-based Chla inversion model, in situ measurements of Chla ($N_{Chla} = 213$) were used as the training dataset (see Section III-C). In addition, 59 in situ water temperature data points were also involved for validation of LST product. These samples were collected during 2017–2021 and spanned the channels of the Yangtze River and the major lakes distributed in the middle–lower Yangtze Plain. The sampling points were selected from different sites in different seasons so that they could represent the varied water quality conditions.

III. METHODOLOGY

A. Data Preprocessing

In this study, the Sentinel-2 L1C images recording the top-of-atmosphere (TOA) reflectance were further processed with atmospheric correction to generate the surface remote sensing reflectance (R_{rs}, sr^{-1}) over water areas. For the retrieval of R_{rs}, the L1C images were processed with the ACOLITE/dark spectrum fitting (DSF) processor (version: 20211124.0) [23]. The DSF algorithm estimates the atmospheric path reflectance based on multiple dark objects in the image scene, without a predefined dark band with negligible water reflectance. This algorithm works well for turbid coastal and inland waters and has been widely used for processing multiband medium-resolution data [24]. The default settings for the MSI sensor were used in this study, and the outputs are the 20-m R_{rs} images. The accuracy of the satellite derived R_{rs} was evaluated using the matched in situ R_{rs} measurements, and the details were presented in the Appendix.

After atmospheric correction processing, the clouds and cloud shadows were detected based on the equation defined with R_{rs} of blue and red bands, i.e., $R_{blue,B_2} - 0.5 \times R_{red,B_4} > 0.08$, and the extracted pixels were removed from the images [25]. The water area was then extracted using the normalized difference water index (NDWI) algorithm, where the default threshold was set to 0 with regional adjustment. In addition, the two MSI pixels adjacent to the river edges were excluded from the detection, to avoid the influence of land adjacency effects. The in situ Chla data were then matched with image pixels with a time interval constrained to <24 h, and the points covered with cloud or fog were filtered out. Considering the highly dynamic characteristics of rivers, most of the in situ Chla data were collected in the lake systems (i.e., $N_{Chla,lake} = 171$ out of the total 213 Chla points).

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TABLE II
HISTORICAL RECORDS OF DIATOM BLOOMS IN THE MIDDLE AND LOWER REACHES OF THE HAN RIVER FROM 2017 TO 2021

Year	Period	River reach	ACD ($\times 10^6$ $cell/L$)	Chla ($\mu g/L$)	Dominant species
2017	Jan. 01-Mar. 31	Huangzhuang–Wuhan	3.0–3.7	11–21	*Stephanodiscus, Cyclotella*
2018	Feb. 08–Mar. 05	Shayang–Wuhan	15.4–34.2	80–100	*Stephanodiscus, Cyclotella*
2019	Jan. 01–Mar. 31	Huangzhuang–Wuhan	2.0–4.0	11.6–18.7	*Cyclotella, Melosira*
2020	Jan. 01–Mar. 31	Huangzhuang–Wuhan	1.7–2.3	5.6–15	*Melosira, Cyclotella*
2021	Jan. 28–Feb. 03	Danjiangkou–Huangzhuang	5.2–8.7	23.1–54.9	*Stephanodiscus, Cyclotella*
	Jan. 16–Feb. 03	Shayang–Wuhan	13.2–41.3	62.1–124	*Stephanodiscus, Cyclotella*

TABLE III
CRITERIA FOR THE STATUS OF DIATOM BLOOMS USED IN THIS STUDY

Grades	Algal cell density ($\times 10^6$ $cell/L$)	Chla concentration ($\mu g/L$)
Non-bloom	0–2	0–10
Mild bloom	2–10	10–50
Severe bloom	>10	>50

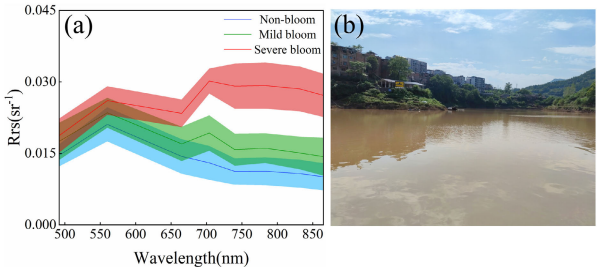


Fig. 2. Feature analysis of diatom blooms. (a) Average R_{rs} spectra extracted from ten multitemporal MSI pixels identified with each status of diatom blooms. (b) Field photograph captured in the Han River in 2021, showing that the water color is yellow-brown during a diatom bloom.

The Sentinel-3 LST product was resampled to a 20-m resolution and registered to match the MSI grids using the Sentinel Applications Platform (SNAP, version: 9.0).

B. Categories of Bloom Status

Generally speaking, the intensities of algal blooms are closely related to the algal density and pigment concentration. However, there are no standard definitions of the threshold values for algal blooms in rivers in the currently reported literature, and these threshold values are likely to vary with the different physical environmental characteristics in different regions. In this study, the status of the diatom blooms was categorized as "nonbloom," "mild bloom," and "severe bloom." In collecting the samples for model training and accuracy validation, the in situ ACD and Chla concentrations from EEMSC were used to evaluate the bloom intensities. The specific criteria are determined based on the historical survey records of diatom blooms in Yangtze River, as listed in Table III [26]. When the algal density exceeds 2.0×10^6/L and the Chla concentration exceeds 10 μg/L, the "mild bloom" status becomes notable, with scattered algae distributed in the water, and the algal particles can be observed in the water. This can be an indicator for the early warning of diatom blooms in river tributaries. When the ACD reaches 1.0×10^7/L and Chla exceeds 50 μg/L, the water surface is likely covered with algal filament or even a thick algal film. In this case, the outbreak of a diatom bloom can be identified, which poses a serious threat to human health.

C. Feature Analysis

1) Spectral Indexes: The primary task for constructing a deep learning-based classification model is to determine the input feature set. For tracking the status of diatom blooms, the spectral signatures for the diatom bloom areas, as distinct from clean water, should be incorporated, and the features that vary with the different diatom bloom statuses need to be considered. However, the spectral resolution of MSI images limits the interpretation of some of the absorption and backscattering features, which can only be observed within the narrow wavelength ranges. In Fig. 2, the average spectral curves extracted from ten multitemporal MSI pixels identified with each diatom bloom status following the instructions in Section III-B are presented, while the average spectrum of clean water pixels is shown for comparison. As shown in Fig. 2, two spectral characteristics can be considered as the main pivotal clues to detect diatom blooms. One is the local reflectance maximum in the visible bands. Due to the strong absorption by chlorophyll pigments, the diatom species exhibit low reflectance in the visible wavelength of <500 nm. The first local reflectance peak for bloom pixels generally occurs at 530–570 nm, dependent on the dominant algal species, while the diatom spectra are shifted to longer wavelengths, relative to the spectra of the blue-green algal species. This can be reflected in the reflectance in the red band of the MSI sensor and results in brown water color in the case of a high ACD. Second, the reflectance peak at ~705 nm in the red-edge channel can effectively distinguish the bloom area from the background. The hyperspectral analysis of phytoplankton species has shown that a unique high backscattering property can be observed for diatoms around this spectral region [27].

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Furthermore, the spectral magnitude varies with the bloom intensities. Therefore, the status of diatom blooms can be identified in terms of both the spectral shape and the reflectance magnitude.

Based on the above analysis, we defined a normalized red-edge-red (NRER) band ratio following the work of [11] and [28] to characterize the spectral magnitude of the local reflectance peak at the red-edge band, which is expressed as:

$$\mathrm{NRER} = \frac{R_{\mathrm{RED-EDGE1},B_5} - R_{\mathrm{red},B_4}}{R_{\mathrm{RED-EDGE1},B_5} + R_{\mathrm{red},B_4}} \tag{1}$$

where the normalized difference between R_{rs} of red-edge ($R_{\mathrm{RED-EDGE1},B_5}$) and red ($R_{\mathrm{red},B_4}$) bands is used to delineate the increased trend of reflectance from the red to red-edge region. The normalized difference vegetation index (NDVI) [29] and FAI [7] were also considered, which have been widely utilized in detecting green-color algal blooms. All three indices are designed to describe the sharp increase in the reflectance spectra at the red-edge region (near 700 nm), which can be commonly observed in the signatures for algal species. However, the peak positions of the spectrum can vary with the different dominant plankton species and bloom statuses. Given this, we used all three feature indices for the model input. The NDVI and FAI are defined as

$$\mathrm{NDVI} = \frac{R_{\mathrm{NIR},B_8} - R_{\mathrm{red},B_4}}{R_{\mathrm{NIR},B_8} + R_{\mathrm{red},B_4}} \tag{2}$$

$$\mathrm{FAI} = R_{\mathrm{NIR},B_8} - R'_{\mathrm{NIR}} \tag{3}$$

$$R'_{\mathrm{NIR}} = R_{\mathrm{red},B_4} + \left(R_{\mathrm{SWIR},B_{11}} - R_{\mathrm{red},B_4}\right) * \frac{\lambda_{\mathrm{NIR}} - \lambda_{\mathrm{red}}}{\lambda_{\mathrm{SWIR}} - \lambda_{\mathrm{red}}} \tag{4}$$

where R_{NIR,B_8} and $R_{\mathrm{SWIR},B_{11}}$ indicate R_{rs} of the NIR and SWIR bands of the MSI sensor, respectively, and λ_{red}, λ_{NIR}, and λ_{SWIR} are the wavelengths for the corresponding bands.

2) Auxiliary Dataset: In addition to the spectral indices directly extracted from the R_{rs} images, the auxiliary information related to the existence of diatom blooms, including water temperature and Chla concentration, was also considered.

The optimum water temperature for the growth of diatom species in the study area is 7 °C–15 °C. *Stephanodiscus hantzschii* can live and grow well even at 2 °C [30]. Therefore, the diatom blooms mostly occur in early spring, which is different from the cyanobacteria blooms generally occurred in summer in freshwater lakes. The water temperature can thus be used as a critical variable to discriminate diatom blooms in terms of environmental factors. The temperature dataset used in this study was the Sentinel-3 LST product resampled to a 20-m resolution (see Sections II-B and III-A). The overall RMSE is 1.31 K, and the average R^2 is 0.958 calculated based on the 59 in situ samples.

Furthermore, Chla concentration is closely related to phytoplankton biomass, which is an important parameter reflecting the intensity of aquatic algal blooms. Therefore, it is feasible to infer the status of a diatom bloom from the regional Chla concentration. To construct the inversion model for Chal estimation, 12 spectral variables of the measured data were taken as the input, i.e., the nine R_{rs} bands of the MSI sensor, band ratio (B_5/B_4), (B_5/B_3), and the NRER reflecting the absorption and scattering features of Chla. In this study, the RF model was selected for the estimation of Chla, which is a model that has been widely used in establishing nonlinear regression relations and has obtained promising results [31]. The model output was the Chla concentration for each image pixel, and the 213 in situ samples were used for the model training (randomly 70% of samples) and validation (30% of samples). In this study, the key parameters of the RF model, i.e., the maximum tree depth (max_depth = 55) and the number of decision trees (n_estimators = 31), were fine-tuned to obtain the optimal accuracy scores. The test accuracies (R^2 = 0.867 and RMSE = 15.18 μg/L) showed that the retrieved Chla was reliable and could be used as auxiliary data in the experiments.

D. Detection of Diatom Bloom Status

1) CNN Model for Bloom Detection: The discrimination of the different statuses of diatom blooms in rivers is essentially a task of classification. In this study, we designed the architecture of the diatom bloom detection model following the structure of the VGG-16 network [32]. VGG-16 is a classical and high-performance CNN-based model for object recognition and image classification, consisting of 13 convolutional layers and three fully connected layers, as shown in Fig. 3.

The input features are made up of 14 variables, i.e., the nine spectral bands of the MSI sensor ($B_2 B_8$, B_{8a}, and B_{11}), the three spectral indices (NRER, NDVI, and FAI), and the two auxiliary variables (Chla and temperature). Given an input patch, it passes through five VGG blocks, which are composed of two or three consecutive convolutional layers and a max-pooling layer. The land pixels and detected invalid pixels (i.e., cloudy pixels) within the patch are masked as 0 and are not included in the calculation and update of the model weights. Each convolutional layer is followed by a rectified linear unit (ReLU) function to increase the nonlinearity [$\mathrm{ReLU}(x) = \max(0, x)$], and every convolutional block is completed with a max-pooling layer to reduce the patch size of the output features. After processing with the max-pooling layer, the model expands the image into a 1-D vector and classifies it through three fully connected layers. Within the fully connected layers, each neuron is connected to all the neurons in the previous layer to calculate the weighted sum, and nonlinear mapping is performed by the ReLU activation function. The final layer is a softmax classifier that translates the output of the convolutional blocks into probabilities for each category. The loss function is cross-entropy loss, which is expressed as

$$L = \frac{1}{N}\sum_i L_i = -\frac{1}{N}\sum_i \sum_{c=1}^{M} y_{ic} \log(p_{ic}). \tag{5}$$

In (5), N indicates the batch size, M is the number of categories, y_{ic} indicates the label of sample i, and p_{ic} indicates the probability of sample i being correctly classified for class c.

2) Implementation Details: The training and validation samples were selected from the satellite images introduced in Section II-B, based on the historical survey records (see Tables II and III) and visual inspection of the reflectance

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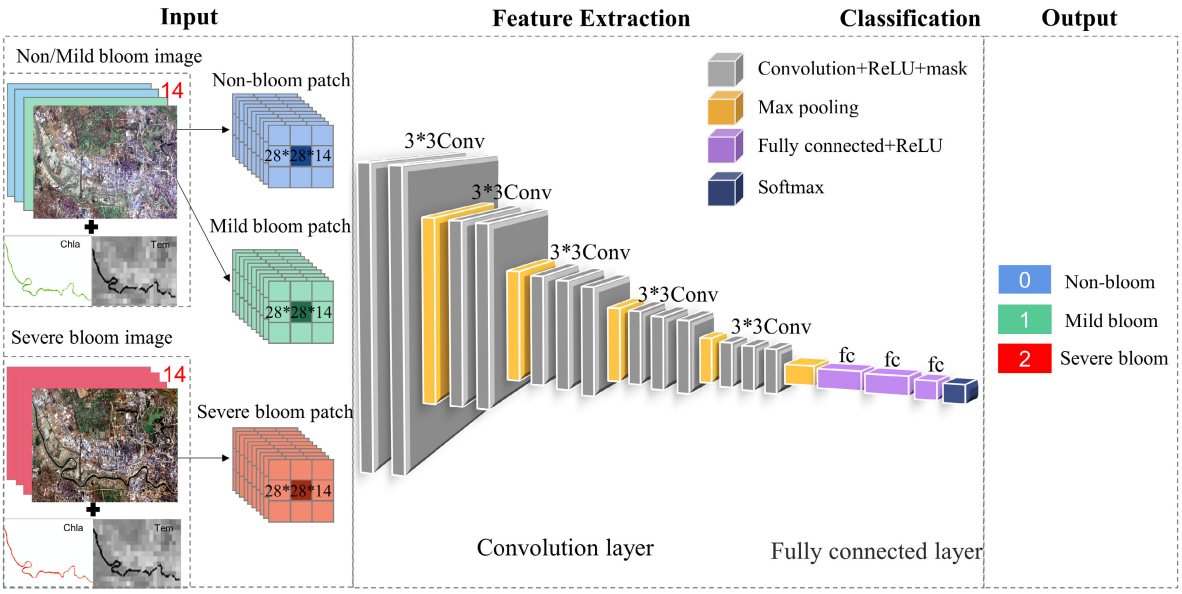


Fig. 3. Structure of the CNN network based on the VGG-16 model used for the detection of the status of diatom blooms to fuse the features of the two branches.

spectra. Some of the historical survey records from EEMSC only provided the average measurements of ACD and Chla for the river sections. Therefore, the survey data were matched with the satellite images, and the samples were selected from the images according to the descriptions of survey records and spectral characteristics of R_{rs}. The selected sample pixels corresponding to different degrees of bloom are then used as central pixels to generate sample patches for the model input. We generated 7663 patches with an input size of 28 × 28, including 2565 nonbloom patches, 2635 mild-bloom patches, and 2463 severe-bloom patches. Among them, 70% of the samples are randomly selected from the 26 images introduced in Section II-B for model training, and the remaining 30% were selected from the independent test images composed of eight images used for accuracy evaluation. Once the model is well trained, the category for the central pixel of the input image patch can be obtained in an automatic manner. In terms of network optimization, the Adam optimizer was used, with the initial learning rate set as 0.0002. The batch size was fixed as 64, and the maximum number of epochs was 100. The code was implemented using the PyTorch framework with an NVIDIA GeForce GTX 1070 GPU.

E. Accuracy Evaluation

Both visual interpretation and quantitative evaluations were performed to evaluate the performance of the diatom bloom detection model. The quantitative accuracy scores were calculated based on the test samples, as introduced in Section III-D. The overall accuracy (OA), user's accuracy (UA), and producer's accuracy (PA) indices are used to report the accuracies, which can be defined as

$$O = \sum_{j=1}^{n} p_{jj} \tag{6}$$

$$U_j = p_{jj}/p_{j+} \tag{7}$$

$$P_j = p_{jj}/p_{+j} \tag{8}$$

where p_{jj} represents the proportion of pixels that were correctly classified for stratum j in the population error matrix, p_{j+} is the proportion of the correctly classified pixels to the total estimated pixels for class j, and p_{+j} is the proportion of the correctly mapped pixels to the corresponding reference map.

IV. Results

A. Visual Interpretation and Evaluation

Fig. 4 shows the detection results for diatom blooms in the middle and lower reaches of the Han River, based on the cloud-free Sentinel-2 images acquired in the early spring during 2017–2020. The results are basically consistent with the survey records, in that mild blooms have been frequently observed between January and March along the Han River in recent years. Moreover, a severe bloom was detected in the river reaches below the station of Shayang in 2018, which indicates a megascopic severe-bloom event spanning ∼360 km.

Zoomed-in views of the multitemporal true-color composite images and the detection results for the three subregions highlighted in Fig. 4 are presented in Fig. 5. Moreover, the spectral curves of the labeled pixels are shown in Fig. 6. Generally, the clean river water is of green color. However, in the case of an outbreak of a severe diatom bloom [see Fig. 5(c), (g), and (k)], the water is tinged yellow-brown, resulting from the brown pigment (i.e., fucoxanthin) existing in diatoms.

The spectral curves [see Fig. 6(b) and (c)] show that R_{rs} of the diatom bloom pixels in the red and red-edge region is much higher than that of the clean water pixels classified as nonbloom. Moreover, the spectra of the diatom bloom pixels are characterized by the two local reflectance peaks at 560 nm (B_3 in green) and 705 nm (B_5 in red edge). The detected mild-bloom pixels [see Fig. 5(a), (e), and (i)] do not show apparent color changes, compared with the nonbloom water pixels. However, the local reflection peak at B_5 centered at

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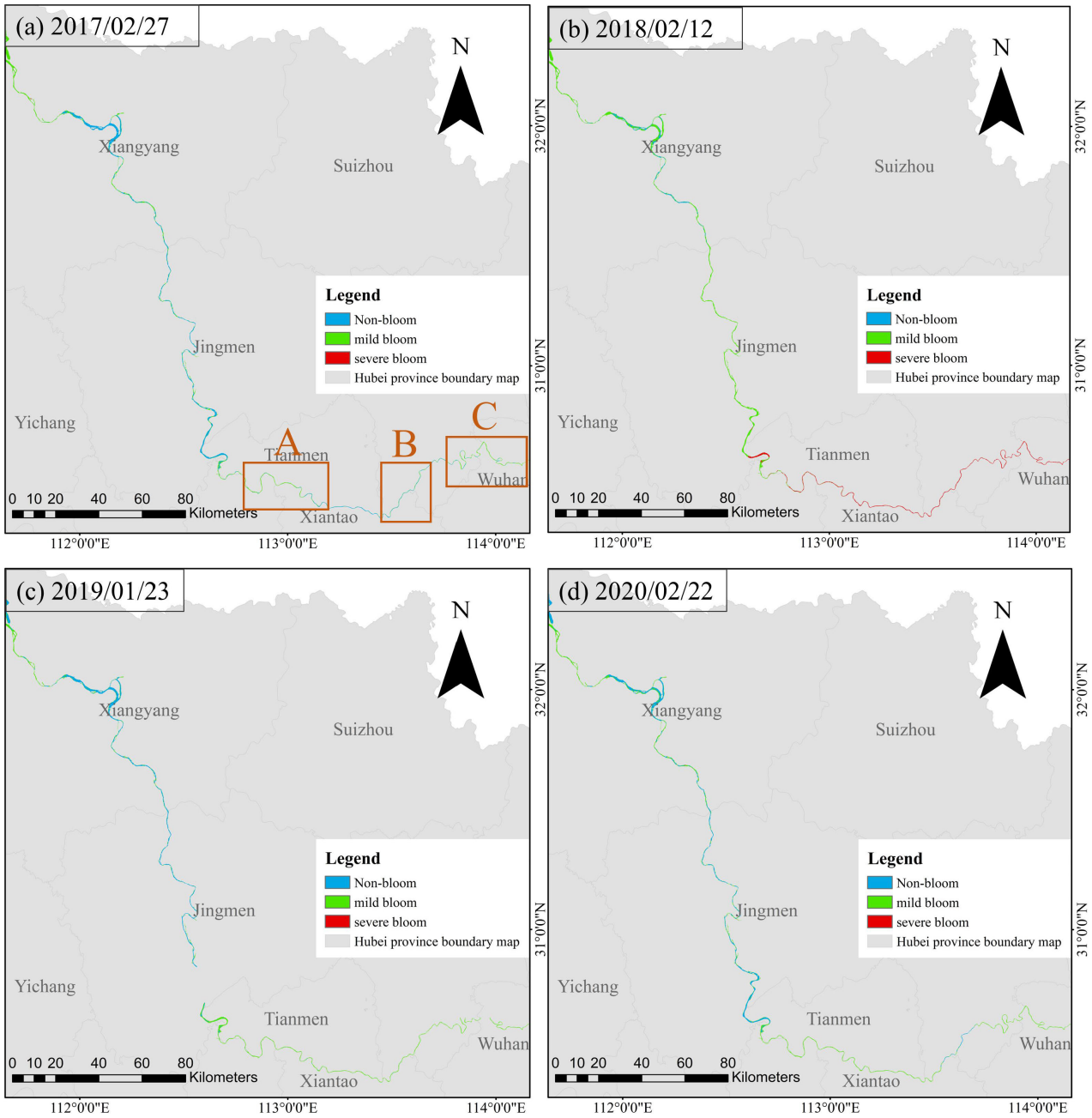


Fig. 4. Detection results for diatom blooms in the middle and lower reaches of the Han River. (a)–(d) Visual maps in early spring from 2017 to 2020. Zoomed-in views of the highlighted regions A–C are given in Fig. 5. The disconnected part between Tianmen and Jingmen in (c) is caused by the clouds covered on the image obtained on 2019/01/23.

705 nm can be used as the pivotal feature to discriminate the status of diatom blooms. Furthermore, the reflection magnitudes provide supplementary information, in that the spectra identified as severe blooms are generally characterized with higher reflectance in B_5 than those of the mild-bloom pixels. The reflectance of B_5 in Fig. 6(c-1)–(c-3) is comparable to that of B_3 in the visible green region, with high concentrations of diatom cells.

B. Correspondence Between Detection Results and Surveying Records

In this section, the results are evaluated in terms of the correspondence between the classification results and the survey records. The in situ survey data collected at Zongguan station between January and March 2021 are used, including Chla and ACD, as shown in Fig. 7. The status of the diatom blooms for the river section at Zongguan station on the six dates with in situ data can be determined according to the criteria defined in Table III.

The detection results in Fig. 8 show that a mild bloom can be detected in the middle of January (2021/01/12), and the diatoms developed rapidly into a severe bloom within a week, where the event was captured by the Sentinel-2 images acquired on 2021/01/19. It can also be observed from the true-color image view shown in Fig. 5(k) that the river water changed into a brown color. This diatom bloom event lasted

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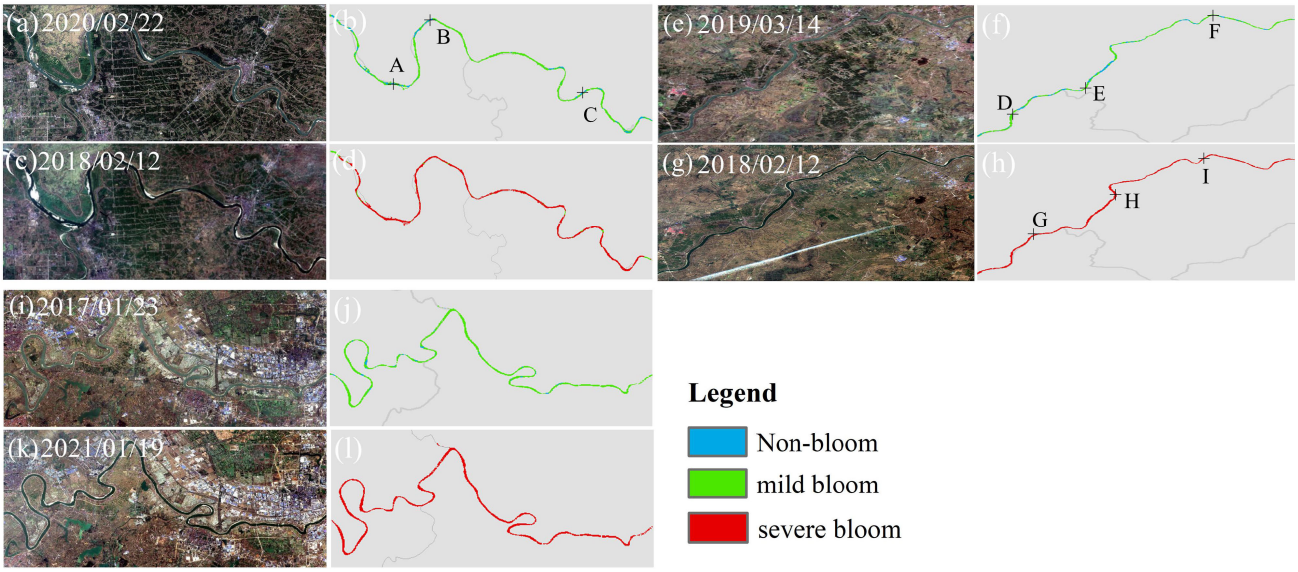


Fig. 5. True-color MSI images (left column) and the corresponding detection results (right column). The spatial extent of the figures corresponds to region A [see (a)–(d)], region B [see (e)–(h)], and region C [see (i)–(l)] highlighted in Fig. 4(a). For each region, the nonbloom or mild-bloom status is used for a comparison with the severe-bloom case. The spectra of the highlighted points in (b), (f), and (h) are shown in Fig. 6.

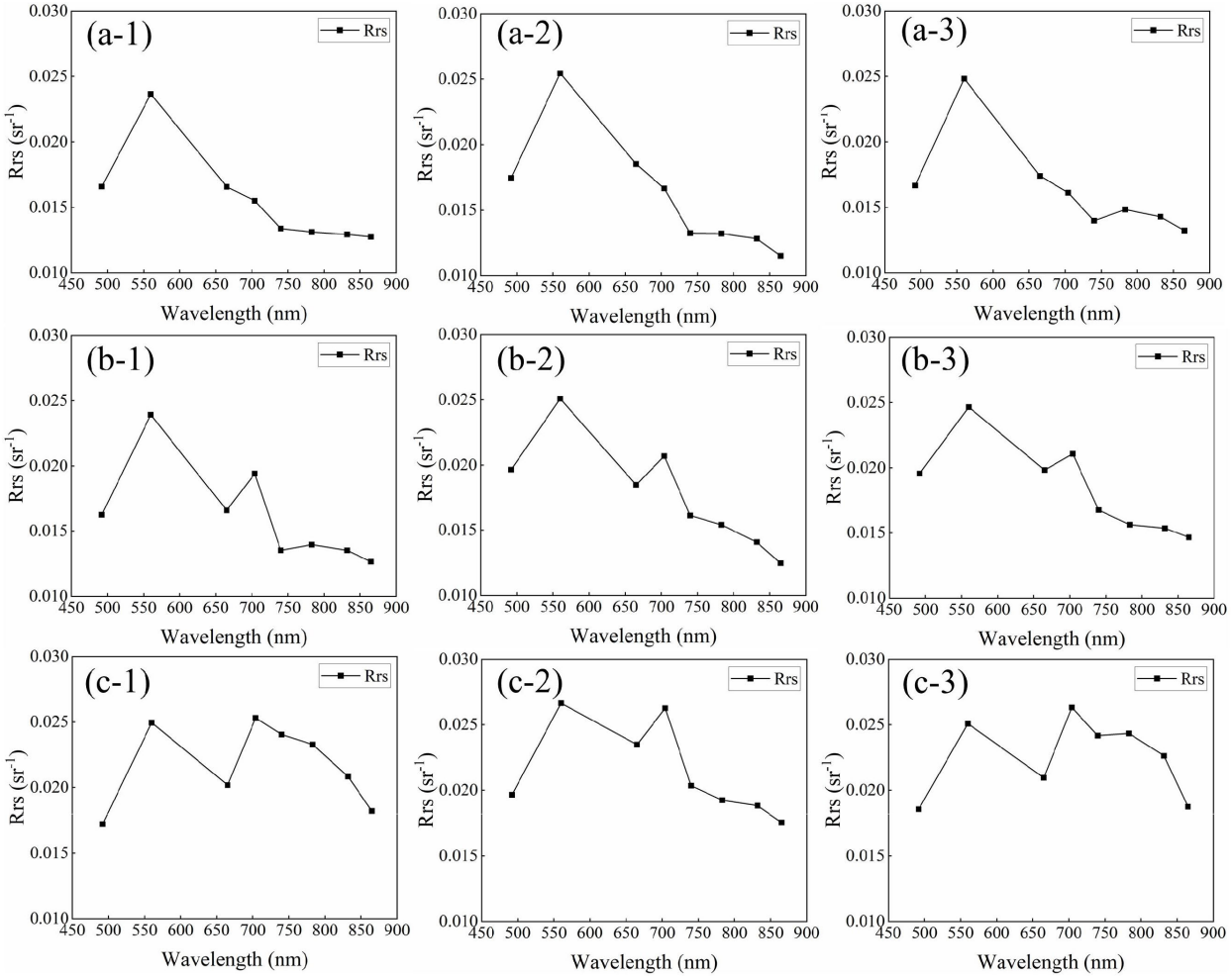


Fig. 6. Satellite derived R_{rs} of the highlighted points in Fig. 5. (a-1)–(a-3) Nonbloom pixels corresponding to points A–C in Fig. 5(b). (b-1)–(b-3) Mild-bloom pixels corresponding to points D–F in Fig. 5(f). (c-1)–(c-3) Spectra of severe-bloom pixels corresponding to points G–I in Fig. 5(h).

for more than ten days, and the water quality changed for the better from early February, which was observed from the monitoring results on 2021/01/29. The diatom bloom dissipated in late February. The mapping results are highly

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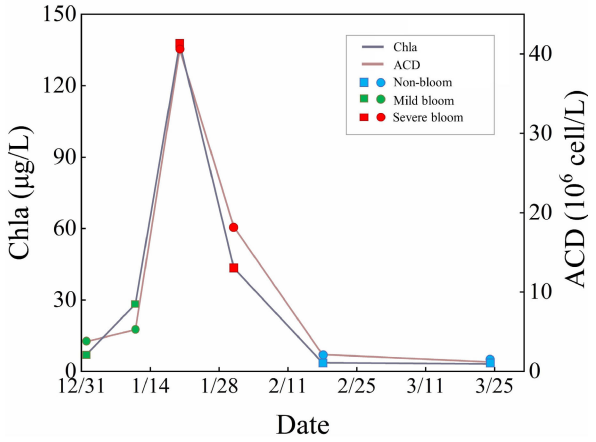


Fig. 7. Records of Chla concentration and ACD at Zongguan station from January to March 2021.

TABLE IV
QUANTITATIVE ACCURACY EVALUATION OF THE CNN AND RF MODEL

Category	CNN OA: 0.927		RF OA: 0.782	
	UA	PA	UA	PA
Non-bloom	0.915	0.923	0.733	0.734
Mild bloom	0.899	0.878	0.737	0.709
Severe bloom	0.976	0.968	0.939	0.950

TABLE V
QUANTITATIVE ACCURACY EVALUATION OF THE DETECTION RESULTS OBTAINED USING THE CNN MODEL WITH DIFFERENT INPUT FEATURE SETS

Category	*RS_Chla_Tem* OA:0.927		*RS_Chla* OA:0.862		*RS_Tem* OA:0.815	
	UA	PA	UA	PA	UA	PA
Non-bloom	0.915	0.923	0.862	0.847	0.799	0.761
Mild bloom	0.899	0.879	0.833	0.769	0.743	0.711
Severe bloom	0.976	0.968	0.932	0.921	0.941	0.906

consistent with the survey records, which demonstrates that the proposed method can generate reliable detection results for the status of diatom blooms in rivers.

C. Quantitative Evaluation and Model Comparison

The quantitative evaluation results are listed in Table IV, including the OA and the accuracy scores (i.e., UA and PA values) for each category. The performance of the RF model is included for a comparison with the CNN-based deep learning model. The quantitative accuracies indicate that the deep learning model outperforms the RF model, where the OA values for the CNN and RF models are 0.927 and 0.782, respectively. Both models show a good performance in detecting the pixels categorized as severe bloom. Comparatively, the task of distinguishing the nonbloom and mild-bloom pixels is more challenging.

Among the results of the deep learning model, the PA value (i.e., 0.899) for the mild-bloom type is relatively low, which indicates that mild-bloom pixels are easily misclassified as the nonbloom or severe-bloom classes. In the early stage of phytoplankton growth, the two-peak feature in the spectral curve for a diatom bloom area is not remarkable and is easily confused with clear water. When the diatom bloom continues to develop, the spectral shapes with the two reflection peaks for the mild-bloom pixels are similar to those of the severe-bloom pixels. When combined with the visual images presented in Figs. 9 and 10, the spectral shapes of the mild-bloom and severe-bloom pixels can be diverse with regard to the reflection intensity at ~705 nm in the red-edge band. The RF model has difficulties in correctly recognizing the mild-bloom status (UA: 0.737 and PA: 0.709). With the good capability of extraction of the high-level and discriminative features, the proposed deep learning method obtains a robust performance in this task.

The proposed method was also compared with the FAI algorithm commonly used for detecting floating algae, and the results are shown in Fig. 11. The FAI images in Fig. 11(a) and (e) were calculated based on the water reflectance, and the classification maps were obtained using an empirical threshold set as 0.005. As shown in Fig. 11(b) and (c), the FAI algorithm cannot distinguish the diatom blooms with different intensities. Moreover, the FAI results wrongly classify the nonbloom pixels as mild-bloom pixels (e.g., Points D and F) in Fig. 11(f) and potentially overestimate the bloom area. Furthermore, compared with the FAI algorithm, which requires manual adjustment of the threshold, the proposed method can obtain the diatom bloom detection results in an automatic manner after model training, and the results can delineate the different statuses of the diatom blooms.

V. DISCUSSION

A. Influence of the Input Features

In this section, we analyze the influence of the input features used to train the CNN model. The relative importance of the 14 input feature elements was calculated, and the ablation tests were conducted by removing each feature element from the input set, as shown in Fig. 12. Among the input features, FAI, Chla, and NRER make the largest contribution to the algal bloom detection, while the single band of B_4 obtains the minimum value of importance. The OA values indicate that the impact of the input features on the results is highly consistent with the relative importance. As the feature indexes are removed one by one from the input set, the resulting accuracy decreases to different degrees. Compared with the OA of 0.927 obtained using the proposed method by incorporating all the 14 feature elements, the differences are negligible when B_4 is excluded in the input (i.e., OA: 0.924). However, given the slight impact on the computational efficiency, we did not exclude B_4 from the input features.

In terms of the contributions of the two environmental variables, the visual example in Fig. 13 shows that the detection results without the involvement of Chla [referring to RS_Tem, Fig. 13(c)] are quite different from the other two cases [see Fig. 13(a) and (b)]. When combined with the spectral curves in Fig. 13(d), it can be seen that the RS_Tem model wrongly categorizes the severe-bloom area as nonbloom (Points A, C, and D). Comparatively, the results obtained by the RS_Chla model are similar to those of the

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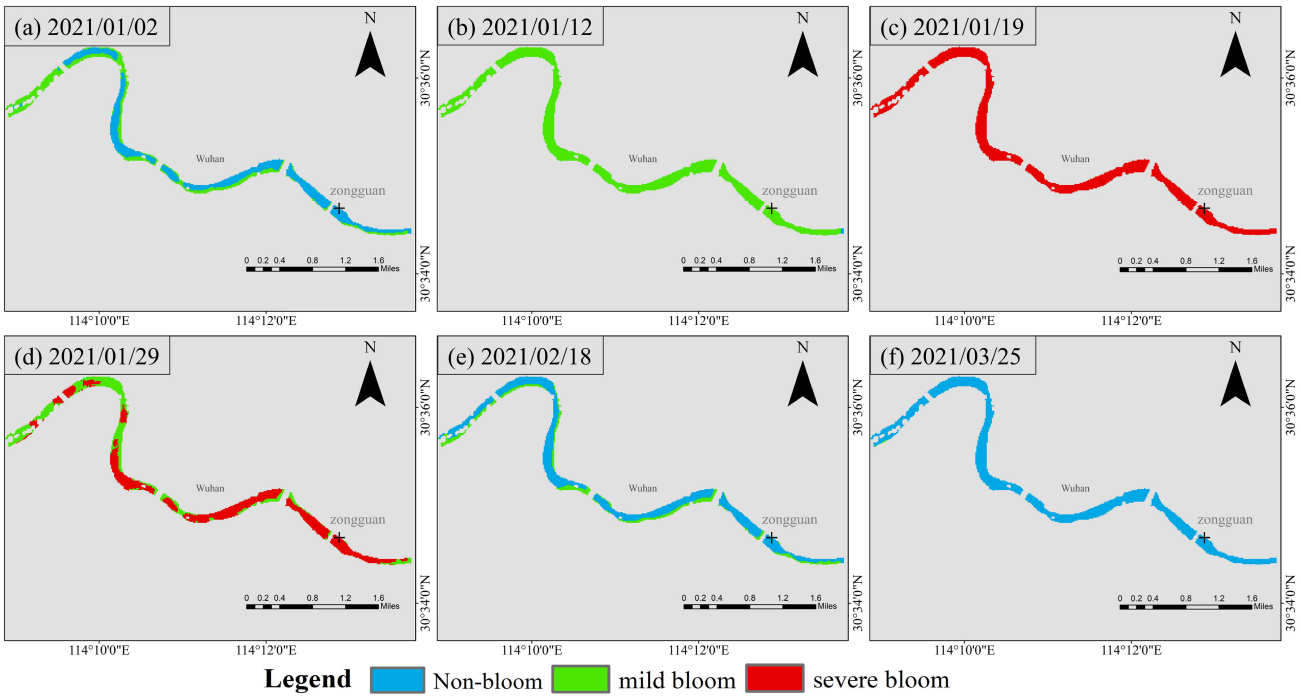


Fig. 8. Detection results for the status of the diatom blooms in the river channel near Zongguan station from January to March 2021. (a)–(f) are the detection results of CNN model at the corresponding date in the Zong Guan station, respectively.

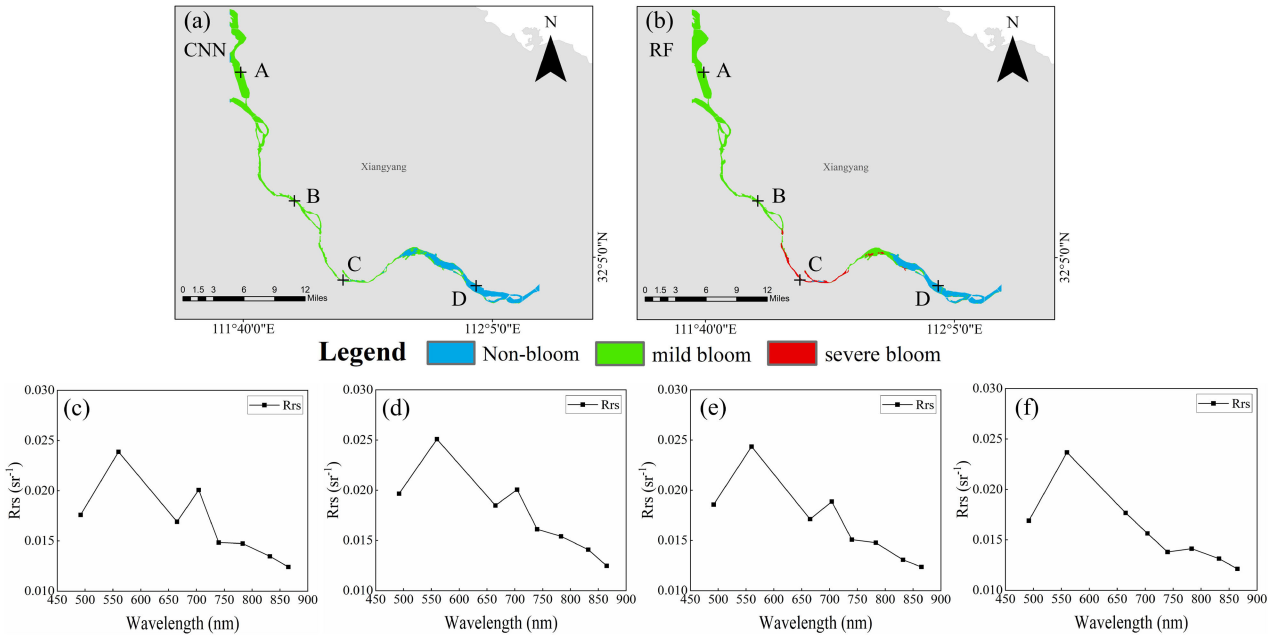


Fig. 9. Comparative performance of the CNN and RF models in diatom bloom detection in the study area on 2017/02/27. (a) and (b) Detection results obtained using the CNN and RF models, respectively. (c)–(f) Corresponding satellite derived R_{rs} spectra of points A–D, which show the spectral characteristics of the mild-bloom status and a typical example of the spectrum of clean water.

proposed method, on the whole, and the detection results for the validation points are consistent with the spectral curves in Fig. 13(d). The OA values for the RS_Chla and RS_Tem models are 0.862 and 0.815, respectively, which represent a 6.5% and 11.2% decline when compared with the OA accuracy of 0.927 for the proposed method. The quantitative results in Fig. 12 and Table V agree with the qualitative analysis, in that Chla has a larger impact on the detection accuracy than water temperature. Furthermore, the detailed results in Table V further indicate that the detection of the nonbloom and mild-bloom categories is more dependent on the two environmental variables.

B. Extensive Applications and Limitations

To validate the generalization ability of the proposed method, the trained CNN model was further tested in the Xiangxi River, which is a large tributary of the Yangtze River located ~40 km upstream of the Three Gorges Dam (TGD).

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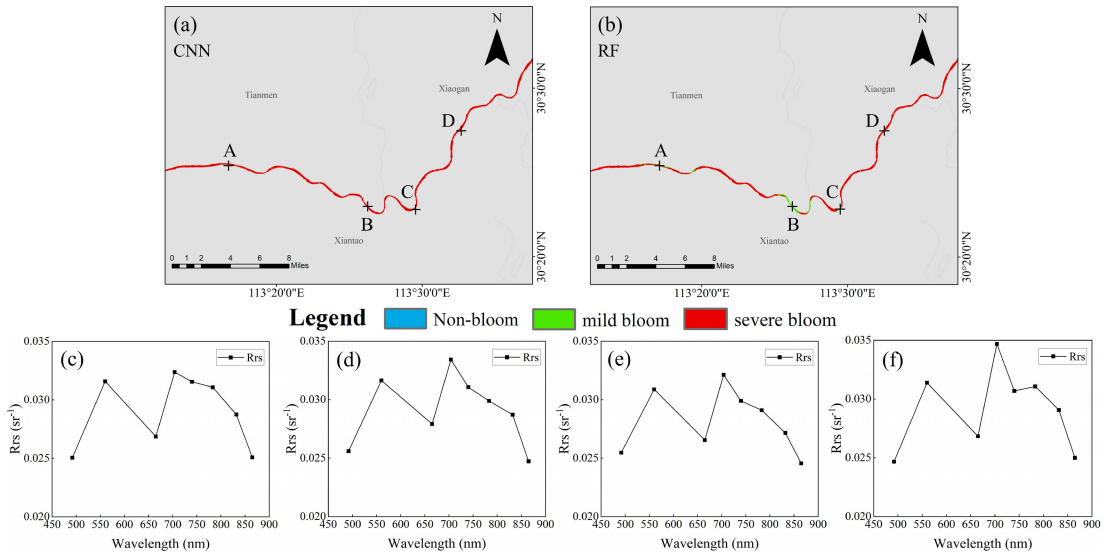


Fig. 10. Comparative performance of the CNN and RF models in diatom bloom detection in the study area on 2018/02/12. (a) and (b) Detection results obtained using the CNN and RF models, respectively. (c)–(f) Corresponding satellite derived R_{rs} spectra of points A–D, which show the distinctive spectral characteristics of the severe-bloom status.

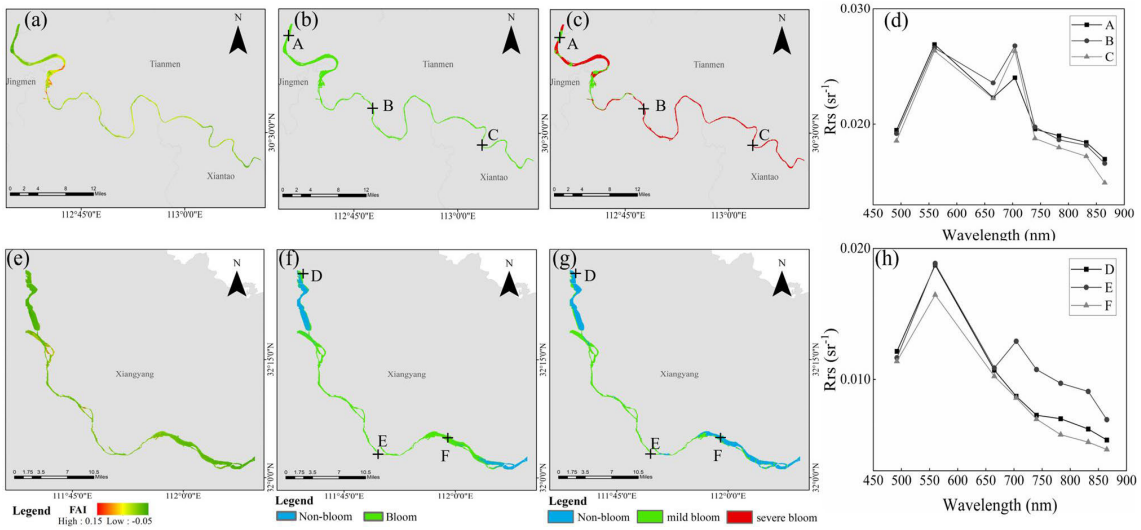


Fig. 11. Comparative performance of the proposed CNN model and the FAI algorithm in diatom bloom detection in the study area on 2018/02/12 (the first row) and 2020/02/22 (the second row). (a) and (e) Calculated FAI image and (b) and (f) classification map obtained using the empirical thresholds for the FAI images. Moreover, (c) and (g) are the detection results of the proposed model. The satellite derived R_{rs} spectra for the sampling points are shown in (d) and (h) for spectral analysis. The legend is shared for the figures of each column.

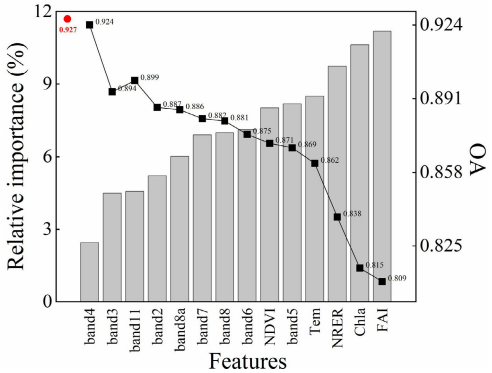


Fig. 12. Relative importance of the feature elements and the obtained OA values by removing the corresponding feature index. The red point in the figure denotes the OA of the proposed method by incorporating all the 14 feature elements.

Eutrophication has frequently occurred in the backwater areas of some of the tributaries, and the diatoms (i.e., *Cyclotella* and *Asterionella formosa*) are the predominant group in this area during the algal bloom period in spring [33].

In this test, the CNN model locally trained using the data collected in the Han River was applied to generate the classification map for the diatom bloom in the Xiangxi River. The results on 2020/04/15 are shown in Fig. 14. For the four points referring to A–D, the two reflectance peaks at ∼550 and ∼705 nm can be observed in the spectral curves. Furthermore, the high reflectance values in the red band at ∼665 nm and the red-edge band at ∼705 nm can be used to distinguish the severe-bloom pixels from the mild-bloom pixels. The detection results are highly consistent with the spectral characteristics. Overall, the results show that the proposed method has a good generalization ability, and the trained model can be used to detect the status of the diatom blooms in the tributaries of the Yangtze River.

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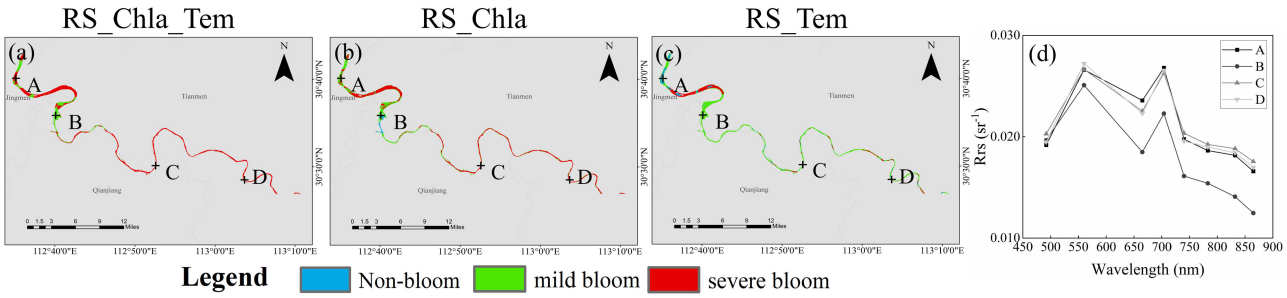


Fig. 13. Comparative performance of the CNN model with different input feature sets. (a) Results obtained using all 14 features as the input, referring to RS_Chla_Tem. The results obtained using the proposed model without water temperature and Chla as the input are displayed in (b) and (c), respectively. (d) Corresponding satellite derived R_{rs} spectra of points A–D.

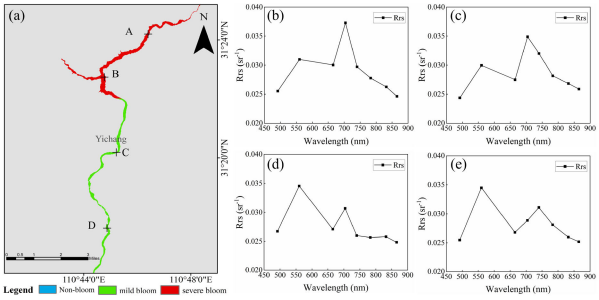


Fig. 14. Detection results obtained in the Xiangxi River using the MSI image captured on 2020/04/15 with the proposed model. (a) Visual map of the bloom detection results. (b)–(e) Satellite derived R_{rs} spectra of points A–D, respectively.

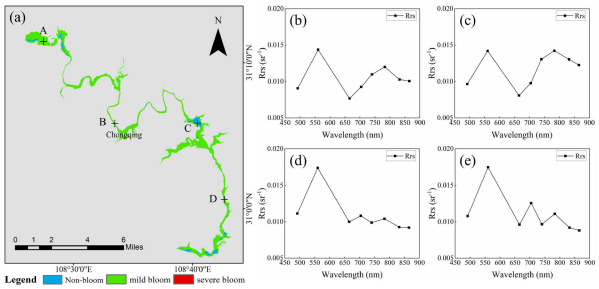


Fig. 15. Detection results for the Xiaojiang River obtained using the MSI image captured on 2020/08/26 with the proposed model. (a) Visual map of the bloom detection results. (b)–(e) Satellite derived R_{rs} spectra of points A–D, respectively.

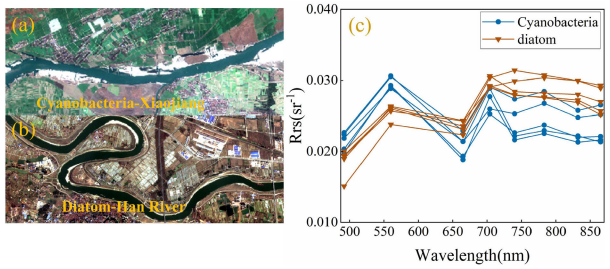


Fig. 16. Analysis of the spectral differences of cyanobacteria and diatom blooms. (a) and (b) Visual examples of water color differences between cyanobacteria and diatom blooms in true-color MSI images acquired on 2020/04/28 and 2021/01/19, respectively. (c) Sampling spectra for cyanobacteria and diatom blooms extracted from (a) and (b), respectively.

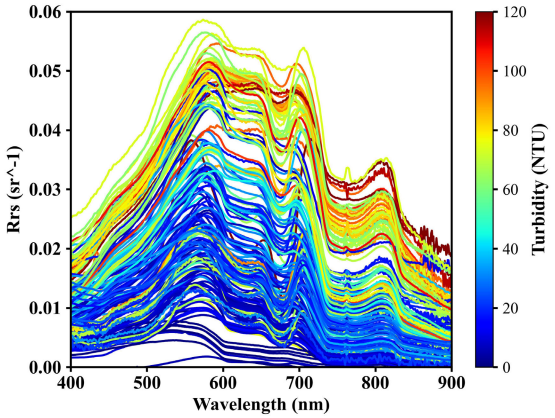


Fig. 17. In situ spectra samples (R_{rs}, $N = 180$) for evaluating ac performance. The colors of the spectral curves varied with the associated turbidity values.

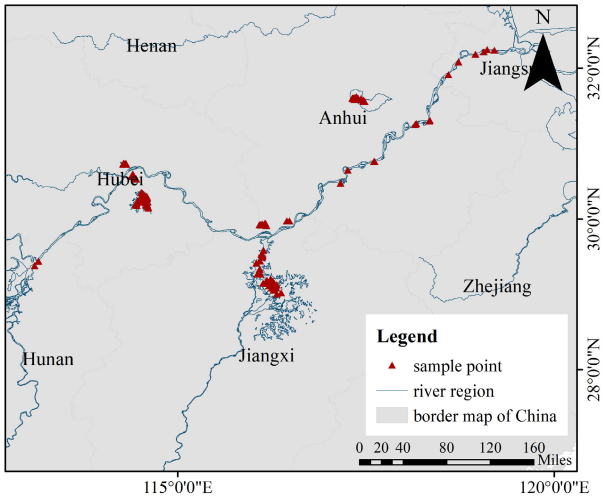


Fig. 18. Distribution of the 180 in situ R_{rs} measurements used for accuracy evaluation of the atmospheric correction results.

The other issue to be discussed is the model's performance in monitoring river channels affected by an outbreak of cyanobacteria bloom. To explore the ability of the proposed model to distinguish between blue–green algae and diatom blooms, a detection test was conducted in the Xiaojiang River using the MSI image acquired on 2020/08/26. The Xiaojiang River is also one of the tributaries of the Yangtze River, where algal blooms are frequently observed during spring and summer. In Fig. 15(a), the detection results show that this area was undergoing a mild diatom bloom. However,

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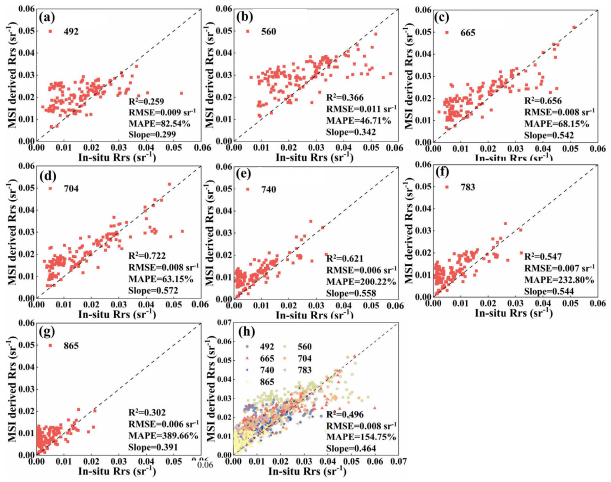


Fig. 19. Evaluation of R_{rs} of MSI images after atmospheric correction with the DSF processor, using the matched in situ R_{rs} data ($N_{R_{rs}} = 180$). (a)–(g) Scatter plots for the seven bands. (h) Overall distribution of the scatter plots and the average evaluation scores.

the field survey records provided by EEMSC indicate that cyanobacteria are the dominant algal species for this algal bloom event, which is also reflected in the spectra for sampling points A–D shown in Fig. 15(b)–(e). From the comparison of the spectra for blue–green algae and diatoms shown in Fig. 16(c), the spectral signatures of MSI sensor are rather similar for the two algal species. However, differences can be observed in the visible bands, where the reflectance of B_3 for blue–green algae is higher than that for diatoms, and the opposite difference is found for the reflectance of B_4. This results in obvious variation in water color for these two categories of algal blooms, as shown in Fig. 16(a) and (b). Overall, the proposed model performs well in identifying the nonbloom and severe-bloom categories. However, it is still challenging to distinguish cyanobacteria blooms from mild diatom blooms, due to the spectral features being easily confused for these two categories. Nevertheless, the algal blooms dominated by cyanobacteria (i.e., *Anabaena flos-aquae* and *Microcystis*) in the tributaries of the Yangtze River are generally also composed of *Cyclotella* belonging to diatoms [34]. From this point of view, the results in Fig. 15 obtained by the proposed model for detection of the status of diatom blooms are still meaningful.

VI. Conclusion

In this study, a CNN model was developed to discriminate the status of diatom blooms based on the diatom bloom intensity using Sentinel-2 MSI images in tributaries of the Yangtze River. The proposed method provides an effective tool to automatically monitor diatom bloom statuses in fluvial systems. The test results obtained in the Han River showed that the proposed method can achieve reliable detection results, in terms of both the visual quality and quantitative validation. The results also showed that the deep learning-based model outperforms the RF model in this task. Specifically, both models can recognize the severe-bloom pixels, while the CNN-based model performs better in accurately distinguishing the nonbloom and mild-bloom areas. Among the input features, both Chla and temperature play significant roles. The locally trained model showed a good generalization ability in the Xiangxi River tributary of the Yangtze River. However, special attention should be paid when applying the model in areas with cyanobacteria blooms because the results might be confusing in the cases with mild blooms, which are difficult to discriminate.

Although only a few works have focused on diatom blooms, the frequently occurring spring blooms have negative impacts and represent a potential risk for the river environment and the safety of drinking water in the riverside cities. The results obtained by the proposed method could provide accurate spatial and temporal information about algal blooms, which is important information for preventing and controlling water pollution. The trained model should be able to effectively inspect diatom bloom occurrences in river tributaries with similar hydrological conditions and environmental systems. Extensive in situ samples and more advanced ac methods can be expected to further improve the model accuracy and the feasibility of the model generalized into national or global scales.

Appendix

The accuracy evaluation has been conducted for the ac results obtained with DSF algorithm using the collected in situ R_{rs}. The in situ R_{rs} measurements were collected using a PSR+3500 (or PSR-1100f) field portable spectroradiometer [Spectral Evolution Inc. (SEI)] following the above water measurement approach [35], [36]. For every sample, the total above-water surface radiance (L_t), downward sky radiance (L_{sky}), and standard reference plaque radiance (L_{pla}) were measured, where viewing angle with an azimuth angle of 135° and a zenith angle of 40° was adopted to avoid the direct sunlight reflected by the water surface [36]. Then, R_{rs} was derived as

$$R_{rs} = \frac{\rho_{pla}\left(L_t - \rho_f \times L_{sky}\right)}{\pi L_{pla}} \tag{9}$$

where ρ_f is the Fresnel reflectance coefficient of the water surface, which is assumed to be 0.022–0.028 in this study [35], [37], and ρ_{pla} is the standard reflectance of the reference panel. For each sampling point, R_{rs} ranging from 400 nm to 900 m with an interval of 1 nm was obtained. Since inland turbid waters do not satisfy the assumption that the reflectance at 750 nm (or longer wavelengths) is zero, the residual reflectance corrections were not conducted for the R_{rs} measurements collected in our study [36]. In Fig. 17, we presented the spectra of the 180 in situ samples used for assessing the performance of atmospheric correction (ac). The in situ data used in this study have also been contributed to the GLORIA dataset, and more details about the methodologies and uncertainties can be referred to [38].

To match up with the valid image pixels, the time difference between the field sampling and satellite transit time should be within ±24 h. Furthermore, a homogeneity test was used to ensure the spatial homogeneity of the water quality around the sampling sites. With the sampling point as the center, the

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sample was regarded as effective when the number of water pixels within the 3×3 window was larger than five, and the coefficient of variation of these water pixels was less than 15% [39]. As a result, there are 180 sampling points involved in the accuracy evaluation, as shown in Fig. 18.

In addition to the commonly used R^2 and RMSE, the MAPE and slope in linear regression are also used for evaluation, and MAPE can be expressed as

$$\text{MAPE} = 100 \times \frac{1}{N} \sum_{i=1}^{N} |\text{est}_i - \text{obs}_i| / \text{obs}_i \tag{10}$$

where obs_i is the ith observed value among the n sampling points and est_i is the ith estimated value.

The evaluation results are shown in Fig. 19 where the band-wise and average scatter plots and accuracy scores were presented. Due to the limited spectral ranges of in situ R_{rs} data, only the validation results for the seven bands from visible to red-edge wavelengths were given. The results showed that the DSF algorithm obtained R_{rs} with the average accuracy scores of R^2 : 0.496 and RMSE : 0.008sr^{-1}. Overall, the DSF exhibited the poorest R_{rs} retrieval at the blue bands, and the accuracies were much better at the red and red-edge bands. The evaluation results of R_{rs} in this study are generally consistent with the previous studies [23], [40], [41], [42].

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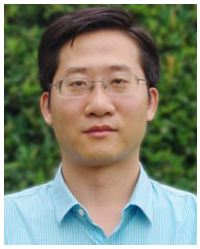
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